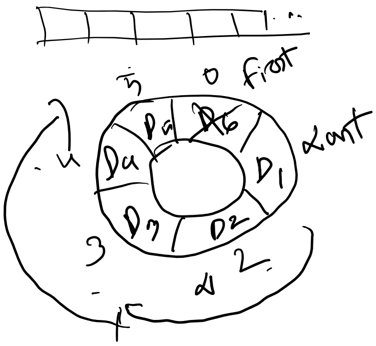
4 Query

Circular Array:

Max-Size = 6



~~Queue~~

Queue

Enqueue (Adding)

Last = (last + 1) % Max-Size

Dequeue

First = (First + 1) % Max-Size

$$\text{last} = (0 + 1) \% 6$$

$$= 1 \% 6$$

$$= \boxed{1}$$

$$\text{last} = (1 + 1) \% 6$$

$$= 2 \% 6 = 2$$

Front()

$\downarrow$

D_1

Dequeue: First = (0 + 1) % 6

$$= 1 \% 6$$

$$= 1$$

$$f(n) = 2n^3 + 100n + 100$$

$$= 2n^3$$

$$O(n) \leq n^3$$

$$n + \log n$$

$\downarrow$

$$n + n \log n$$

$$n + \quad n +$$

Big-O (order of n)

O(n) $5n^3 + 2n^2 + 100n + 200$

40 $16 + 200 + 200$ $n = 2$

$5 \times 1000^3 + 2 \times 1000^2 + 200$

$+ 200$

$5n^3 + 2n^2$

$n = 100$

n^3

$5n^3$

n^3

$O(n) = n^3$